

Data Analysis and Modeling Report

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1. Introduction

The objective of this assignment is to build a predictive model for a continuous target variable y using a set of numeric and categorical features. Rather than assuming a single homogeneous relationship across all data, this study adopts a mode-wise modeling strategy, where separate models are constructed for each operational mode. This decision is guided by exploratory visualizations, which reveal substantial heterogeneity in the relationship between the feature x and the target variable y across different modes. Model performance is evaluated using Root Mean Squared Error (RMSE), and residual diagnostics are used to assess calibration and error structure.

2. Data Preprocessing and Feature Engineering

Several preprocessing steps were applied prior to modeling. First, non-informative and unstructured fields ("row_id" and "note") were removed. The dataset contains 130 missing values in note and 9 missing labels in y . I remove samples with missing y to avoid label-missing bias in supervised training.

The timestamp feature was parsed into datetime format, and invalid timestamps were removed. From the valid timestamps, interpretable temporal features (hour, day, weekday, and month) were extracted to capture potential time-related patterns. The original timestamp column was dropped afterward to avoid redundancy.

Categorical variables ("mode", "category", and "sensor_id") were converted to string type to avoid imposing artificial ordinal structure. For modeling, the primary numeric features used were x , hour, day, weekday, and month. This preprocessing pipeline ensured that the dataset was clean, structured, and suitable for regression modeling.

3. Exploratory Data Analysis and Visualizations

Figure 1 shows the distribution of the target variable y and feature x . The distribution of y is unimodal with moderate spread and no extreme skewness, indicating that transformation of the target variable is not strictly necessary. This suggests that standard regression methods can be applied on the original scale of y . The non-uniform distribution of x

suggests that the relationship between x and y may vary across different regions of the feature space, motivating the use of nonlinear models for some subsets of the data.

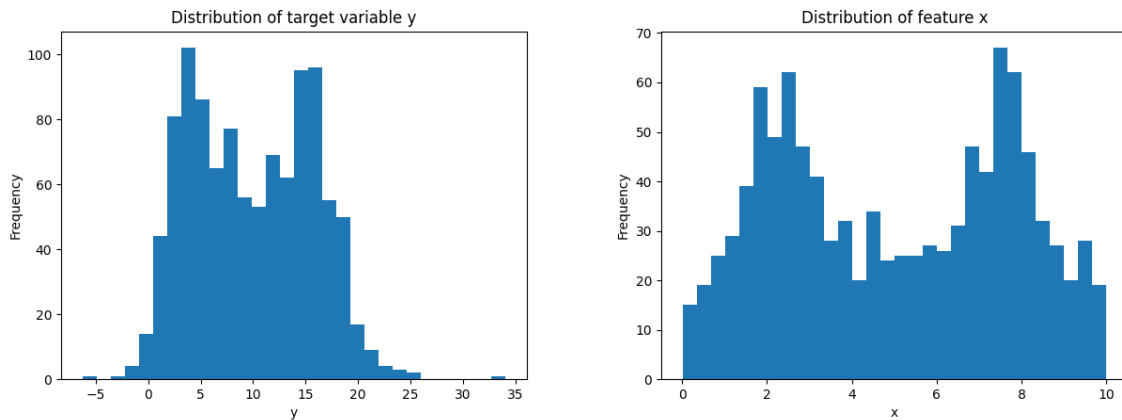


Figure 1 Distribution of X & Y

Figure 2 visualizes the scatter plot of x versus y , grouped by mode. This figure reveals that different modes follow markedly different patterns. Some modes exhibit approximately linear relationships, while others show curved or clustered structures. This visualization provided the primary motivation for adopting a mode-wise modeling strategy, rather than fitting a single global model to all data.

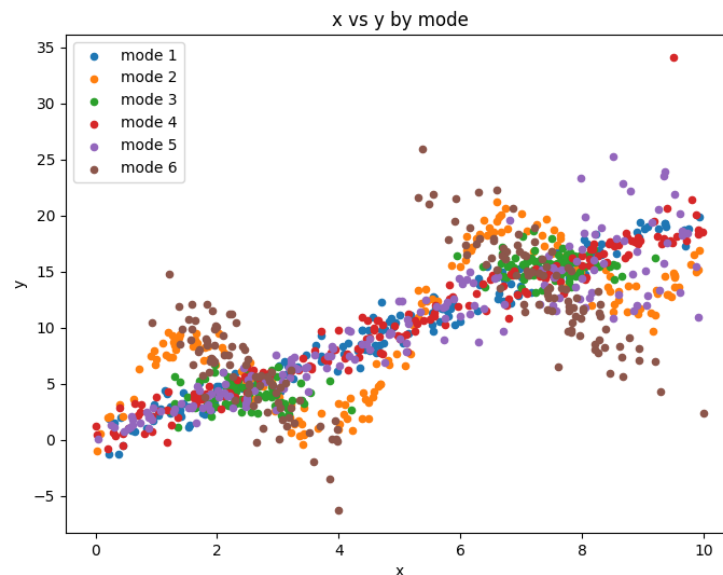


Figure 2 XY Plot

Figures 3 further decompose the x - y relationship by individual mode. For modes 1 and 4, the scatter plots indicate roughly linear trends with relatively low dispersion, suggesting that linear regression is appropriate. Mode 2 exhibits a smooth nonlinear pattern,

motivating the use of polynomial regression. Mode 3 shows two visually separable clusters rather than a continuous functional relationship, indicating the presence of latent subgroups; therefore, a clustering-based approach was used. Mode 5 displays higher noise and mild curvature, making it more difficult to model accurately. Mode 6 shows different patterns across categories within the same mode, motivating a further split by category before modeling.

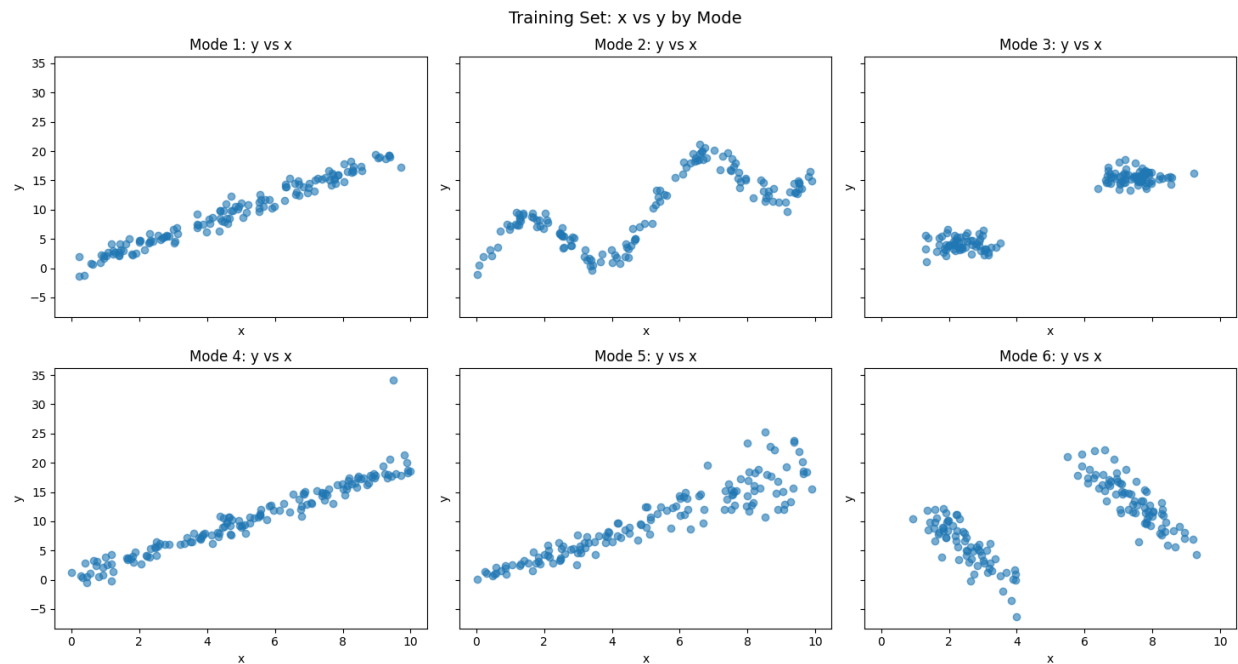


Figure 3 XY Plots by Mode

4. Modeling Strategy

A single global regression model was not adopted because exploratory analysis clearly showed that the functional relationship between x and y varies substantially across modes. Fitting a global model would force a single functional form to approximate multiple incompatible patterns, likely leading to systematic bias in some modes.

Therefore, I grouped the data by mode and built separate models for each group, because the relationship between x and y exhibits distinct patterns across different modes based on visual inspection. This mode-wise modeling strategy reduces model bias by allowing each sub model to better align with the local structure of the data.

For modes 1 and 4, linear regression was used due to the approximately linear relationship between predictors and the target variable. For mode 2, polynomial regression was applied to capture nonlinear trends in x. Mode 3 was modeled using KMeans clustering on x, followed by predicting the mean target value within each cluster, effectively approximating

a piecewise constant model. Mode 5 was modeled using low-degree polynomial regression to capture mild curvature while limiting overfitting. Mode 6 was split by category, and separate linear regression models were trained for each subgroup to account for category-specific behavior.

For Mode 3, the scatter plot suggests the presence of two separated clusters with relatively flat internal structure. An initial attempt was made to combine K-means clustering with local linear regression models fitted within each cluster. However, this approach resulted in a higher validation RMSE (1.702) compared to a simpler cluster-mean baseline model.

This outcome suggests that within each cluster, the relationship between x and y is weak or approximately constant and fitting a regression model introduces unnecessary variance without significantly reducing bias. As a result, the simpler cluster-mean model was selected as the final modeling approach for Mode 3, as it achieved better generalization performance.

Overall, this modeling strategy emphasizes alignment between visual evidence and model choice, ensuring that each model reflects the structural properties observed in the data.

5. Model Evaluation and RMSE by Mode

Validation performance was evaluated using RMSE for each mode. The results indicate that simpler and more structured modes achieve lower error, whereas noisier or more heterogeneous modes exhibit higher RMSE. In particular, Mode 1 (RMSE ≈ 1.01), Mode 2 (RMSE ≈ 1.21), Mode 3 (RMSE ≈ 1.01), and Mode 4 (RMSE ≈ 1.07) demonstrate relatively strong predictive performance, suggesting that the chosen models capture the dominant patterns in these regimes effectively.

In contrast, Mode 5 (RMSE ≈ 2.19) exhibits substantially higher error, reflecting the higher variance and weaker structural relationship observed in its scatter plot. This suggests a relatively low signal-to-noise ratio in Mode 5, which limits achievable predictive accuracy regardless of model choice. Mode 6 (RMSE ≈ 1.79) also shows elevated RMSE, consistent with its internal heterogeneity across categories.

These results confirm that data heterogeneity across modes directly translates into differences in predictive difficulty, and that a single global model would struggle to achieve uniformly good performance across all regimes.

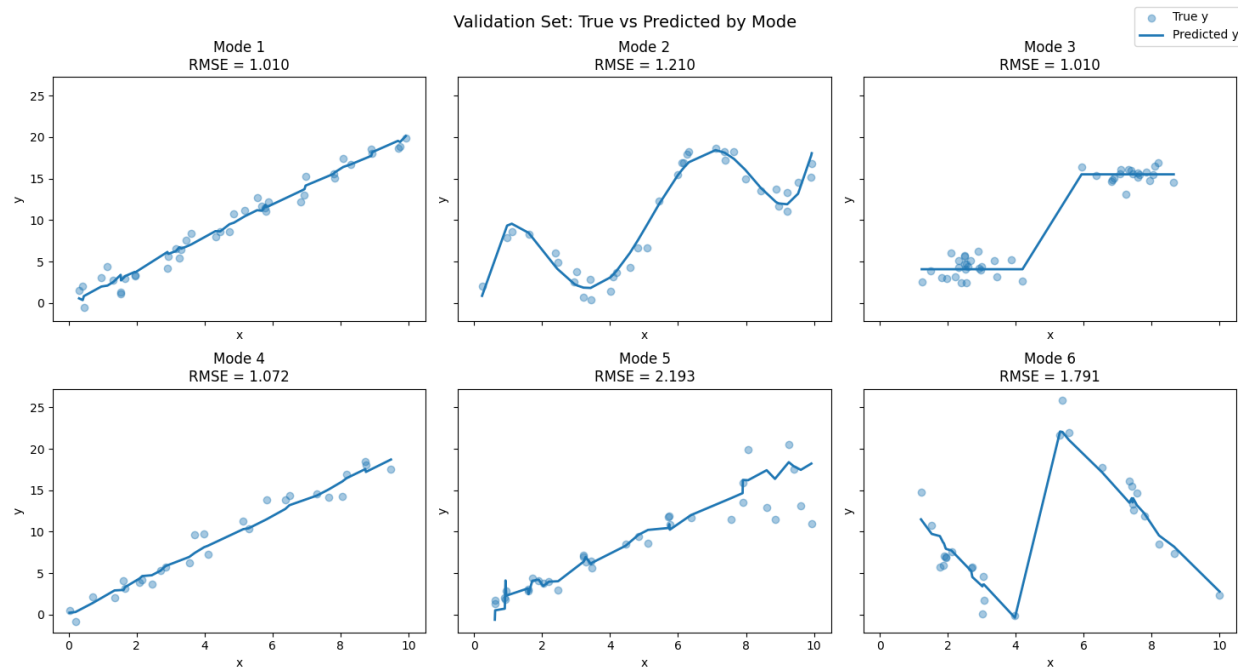


Figure 4 True & Predict Plots by Mode

6. Residual Diagnostics

Figure 5 presents the overall residuals versus predicted values. Residuals are generally centered around zero, suggesting that the models are approximately unbiased. However, larger variance is observed for higher predicted values, indicating mild heteroscedasticity.

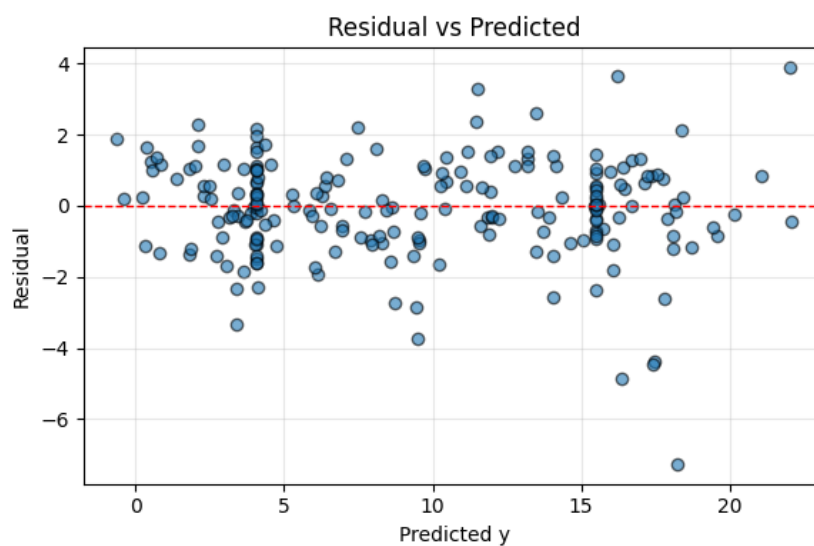


Figure 5 Residual Plot

Figure 6 shows residual plots by mode. Modes with simpler structure (such as modes 1 and 4) exhibit tighter residual distributions, while modes 5 and 6 show larger dispersion, confirming that these modes are inherently more difficult to predict.

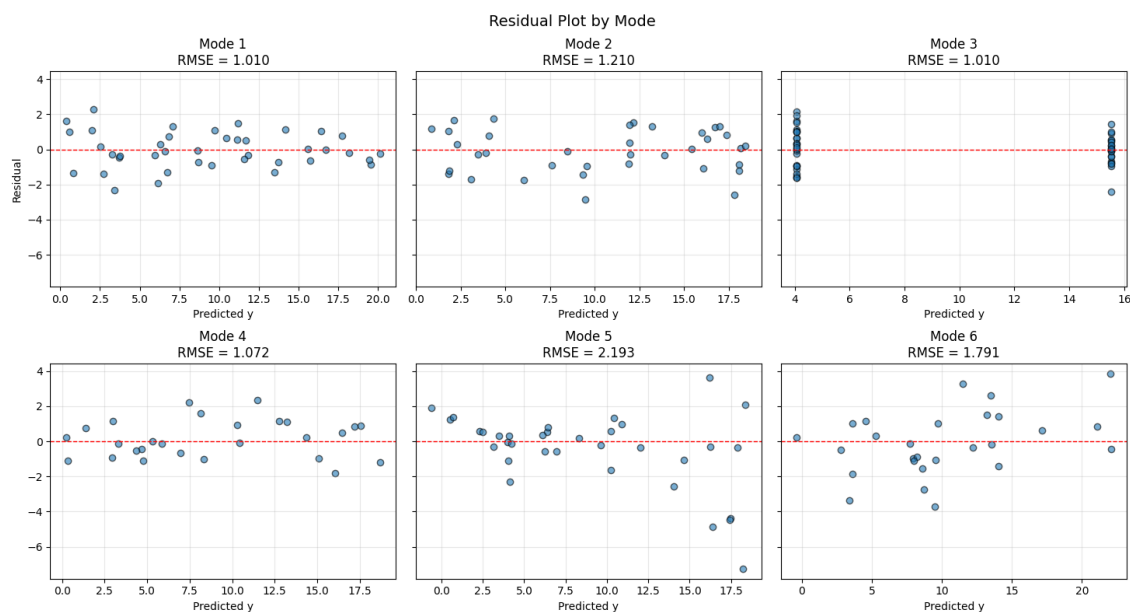


Figure 6 Residual Plots by mode

Figure 7 presents the overall residual distribution, which is approximately symmetric but exhibits heavier tails. This suggests that while most predictions are accurate, occasional large errors remain, particularly in noisy regimes.

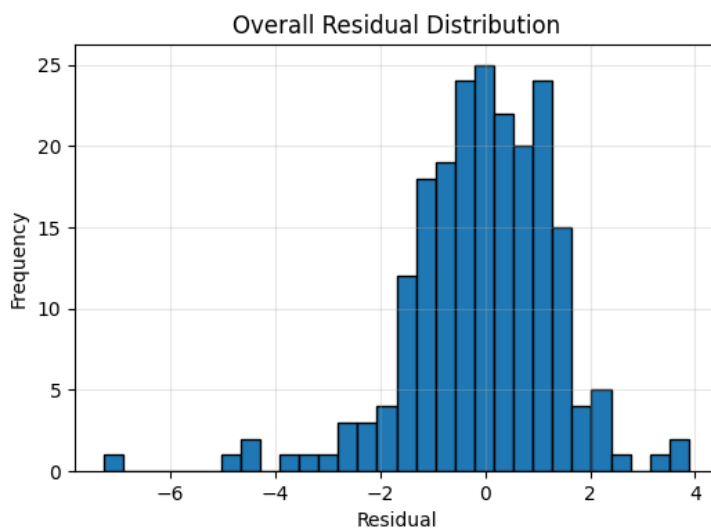


Figure 7 Residual Distribution

7. Estimation of Test Set Error

The validation RMSE by mode provides a reasonable estimate of expected test-set performance. Because the model is trained separately for each mode, the overall test error can be interpreted as a weighted combination of mode-specific errors. Modes with higher RMSE (notably modes 5 and 6) are expected to dominate the overall prediction error if they constitute a substantial portion of the test data.

Given the stability of RMSE across validation and the alignment between visual diagnostics and model choice, the validation RMSE is considered a credible estimate of out-of-sample performance. Nevertheless, the presence of noisy modes implies that some irreducible error remains.

8. Modeling Challenges and Limitations

Several challenges were encountered during modeling. First, strong heterogeneity across modes required a departure from conventional global modeling approaches. Second, some modes exhibited limited structure and high noise, constraining achievable accuracy regardless of model choice. Third, the clustering-based approximation used for mode 3 simplifies potentially more complex substructures and may underfit finer-grained patterns.

Despite these limitations, the chosen approach prioritizes interpretability and alignment with data-driven evidence, ensuring that modeling decisions are transparent and justifiable.

9. LLM Usage & Individual Contribution

I used the large language model (LLM) primarily to generate and compare possible modeling strategies. The LLM suggested alternative approaches such as linear regression, polynomial regression, clustering-based methods, and subgroup modeling for specific modes. Example prompts included: “What modeling approaches are suitable when different modes may follow different functional forms?” and “When should polynomial regression be preferred over linear regression?”

To ensure correctness, all LLM suggestions were verified by inspecting actual dataset values and data types, manually reviewing suggested code, and running the corrected code to confirm that group assignments and RMSE calculations were accurate.

This assignment was independently by Yuting Liang (B01010572). I conducted all stages of the assignment, including data cleaning and exploratory data analysis, feature engineering, per-mode modeling, validation and error analysis, visualization, and report writing.

10. Conclusion

This assignment demonstrates a complete workflow from preprocessing and exploratory analysis to mode-specific modeling and residual diagnostics. Visual exploration revealed that the relationship between features and the target variable varies substantially across modes. By explicitly grouping data by mode and tailoring model choice to observed patterns, the final modeling framework achieves strong performance for structured regimes while transparently highlighting the limitations imposed by noisy and heterogeneous modes. This approach emphasizes that effective predictive modeling is not solely about algorithm choice, but about aligning modeling assumptions with empirical data structure.